

Sam McClenaghan

780-221-1327 | sam@aream.ca | [LinkedIn](#) | [GitHub](#) | smccl.ca

EDUCATION

University of Victoria

BSc (Honours) in Computer Science, Minor in Mathematics

Coursework: Operating Systems, Computer Organization, Data Structures, Signals & Networks

Victoria, BC

Sept 2023 – Present

EXPERIENCE

Software Engineer, Intern

Vertical City

May 2026 – Present

Victoria, BC

- Shipped an in-house video ad pipeline across AWS (S3, MediaConvert, CloudFront) to deliver millions of videos a day, replacing a paid third-party service.
- Cut vendor bandwidth costs by 90%, moving delivery in-house and compressing each ad 5x with VMAF-guided encoder tuning.
- Built a perceptual-hashing system using ffmpeg and pHash to detect duplicate ads that survive re-encoding and slip past exact file hashing.

Full Stack Developer, Intern

Leanpub

May 2025 – Dec 2025

Remote (Victoria, BC)

- Rebuilt Dropbox API integration to streamline file sync and publishing flows for authors using AWS S3 and background job queues.
- Implemented end-to-end background job progress tracking using Redis and a progress bar with reliable completion/failure handling.
- Hardened critical production flows in a large-scale Rails monolith with RSpec and FactoryBot test coverage.

Software Developer, Intern

Trust Science

Nov 2022 – Aug 2024

Edmonton, AB

- Developed and deployed a file version-control system using HTTP/2 and JSON Patch to enable delta storage, reducing AWS S3 costs by 15%.
- Engineered a real-time financial analytics platform using DynamoDB and Elasticsearch, improving query performance by 50%.
- Modernized front-end architecture with Vue 3 Composition API and serverless AWS Lambda services, accelerating partner onboarding by 45%.

PROJECTS

CourseFlow | *Go, React, MySQL, Docker, DigitalOcean*

2025 – 2026

- Built CourseFlow, a course-planning web app used by thousands of UVic students, with course search, live enrollment data, interactive timetable building, and shareable schedules.
- Designed a Go and MySQL backend that ingests catalog and live-enrollment data, then computes conflict-free lecture, lab, and tutorial combinations for the React scheduler.

Anchor | *Next.js, TypeScript, Supabase, Anthropic SDK, Resend*

2025

- Built a biometric mental-health companion at the UVicHacks × Claude hackathon (team of 4) that detects distress from HR/HRV data and provides Claude-powered grounding support.
- Owned backend and DevOps: designed Supabase/Postgres migrations, deployed with Vercel, and implemented Garmin data ingestion, normalization, distress detection, and episode logging.

TECHNICAL SKILLS

Languages: Ruby, Go, JavaScript/TypeScript, Java, Python, C, SQL (MySQL), HTML/CSS

Frameworks & Libraries: Ruby on Rails, RSpec, Minitest, React, Next.js, Remix, Tailwind CSS

Developer Tools: Git, Docker, Jira, AWS (Lambda, S3, DynamoDB), CI/CD